Module 08: Planning — Ethics, Values, and the Normative Dimension

> **Course**: Active Inference 401 | **Unit**: Philosophical Foundations | **Audience**: Advanced undergraduates / graduate students

Learning Objectives

1. Analyze the **normative implications** of Active Inference — can an "is" yield an "ought"?
2. Evaluate the relationship between **preferences (C vector)**, **values**, and **ethics** within the framework.
3. Examine Active Inference's implications for **AI ethics**, **well-being**, and **the good life**.

Introduction

Active Inference describes how agents *do* behave — they minimize free energy. But does this descriptive framework carry *normative* implications? Can it tell us how agents *should* behave? This module investigates the philosophical territory where free energy meets ethics, values, and human flourishing. This is among the most contested and philosophically charged areas in Active Inference research.

Key Concepts

1. The Is-Ought Problem and Active Inference

Hume's guillotine: You cannot derive an "ought" from an "is." If the FEP describes what organisms *do*, it cannot directly prescribe what they *should* do.

However, Active Inference offers a subtle relationship between description and prescription:

The descriptive claim: All self-organizing systems minimize free energy — this is a physical fact about nonequilibrium steady states.

The quasi-normative claim: *Given* an agent's preferences (C vector) and model structure, there are *better* and *worse* policies — those that minimize vs. increase expected free energy. This is normative *relative to* the agent's generative model. The norms are hypothetical imperatives ("If you want X, then you should do Y") rather than categorical imperatives.

The deep question: Where do preferences come from? If the C vector is set by evolution, development, and learning, then "values" are ultimately describtive — they're the states the organism is adapted to expect and maintain. If values are *constituted* by the generative model rather than discovered by it, the is-ought gap may be bridged at the level of autopoietic selfhood: to be the kind of system that you are *is* to value certain states.

2. Preferences, Values, and the Good Life

The C vector — the agent's preferred observations — is the formal representation of values in Active Inference. This raises profound questions:

Are all preferences equal? The framework itself is value-neutral — it doesn't distinguish between a C vector encoding altruism and one encoding sadism. The math works the same either way. This means Active Inference, by itself, cannot ground categorical ethical claims.

Can preferences be evaluated? Several approaches emerge:

- **Evolutionary fitness**: Preferences shaped by natural selection tend to be adaptive — they keep the organism alive. But adaptive ≠ ethical (aggression can be adaptive).
- **Flourishing (eudaimonia)**: Following Aristotle, "good" preferences are those that promote long-term flourishing — allostatic integrity across many timescales. This connects to the FEP's emphasis on homeostatic and allostatic regulation.
- **Coherence**: Preferences that are internally consistent and well-calibrated to reality (high model evidence) might be considered "better" — they lead to sustainable, low-free-energy existence. Incoherent preferences generate unnecessary prediction errors.
- **Intersubjective alignment**: Preferences that facilitate mutual model alignment and social coordination might be considered prosocial — they reduce collective free energy.

The hedonic vs. eudaimonic distinction: Simple free energy minimization might seem to favor hedonic comfort (the "dark room"). But EFE includes epistemic value — the drive to reduce uncertainty. A system that is truly minimizing expected free energy over long time horizons must be curious, exploratory, and growth-oriented — closer to eudaimonia than hedonism.

3. AI Ethics and Active Inference

If artificial agents can be designed as Active Inference systems, ethical questions become urgent:

Value alignment: How do we set the C vector of an AI system to align with human values? This is the formal version of the AI alignment problem. Active Inference provides a framework — set C = human-compatible preferences — but doesn't solve the problem of *what* those preferences should be.

Autonomy vs. control: An Active Inference agent, by definition, acts to maintain its own generative model. This creates a tension: we want AI systems to serve human goals (external C vector), but the agent's own dynamics may develop internal preferences through learning. How do we ensure these remain aligned?

Transparency: Active Inference agents have interpretable generative models — you can inspect their A, B, C, D matrices and understand *why* they take specific actions. This is a significant ethical advantage over black-box deep learning systems.

The digital agent's moral status: If an AI system meets the criteria for Active Inference agency (Markov Blanket, generative model, free energy minimization), does it have moral status? The graduated ontology from Module 02 suggests moral status might come in degrees — but this remains philosophically unresolved.

4. Responsibility and Moral Agency

Active Inference reframes responsibility:

Responsibility as model ownership: An agent is responsible for its actions to the extent that those actions flow from its own generative model (not external coercion). This is the compatibilist account applied to ethics.

Moral development: The development of moral reasoning can be modeled as the progressive refinement of a generative model that includes social and ethical priors. A morally mature agent has a model that predicts and values the consequences of its actions for others.

Moral emotions: Guilt, shame, empathy, and moral indignation can be understood as prediction errors within the social-moral generative model. Guilt arises when one's actions violate one's own moral priors (self-model prediction error). Shame arises when one's actions violate perceived social expectations (social model prediction error).

5. Political Philosophy and Collective Free Energy

Active Inference has implications for political philosophy:

Justice as model alignment: A just society is one where individuals' generative models are sufficiently aligned to enable cooperation without coercion — collective free energy is minimized through voluntary coordination rather than imposed order.

Power as precision control: Political power can be understood as the ability to set the precision of others' prediction errors — to control what information is attended to and what is ignored. Propaganda = precision manipulation of collective generative models.

Democracy as collective inference: Democratic deliberation can be modeled as multi-agent belief updating — citizens exchange messages (arguments, evidence, narratives) and converge (or fail to converge) toward shared beliefs about collective action.

Summary

Active Inference doesn't directly solve ethics — it cannot derive "ought" from "is." But it provides a formal framework for understanding values (C vector), evaluating preference coherence, analyzing moral development, and addressing AI alignment. The framework suggests that the good life involves coherent, well-calibrated preferences that balance hedonic comfort with epistemic growth, and that ethical maturity involves expanding one's generative model to include the welfare of others.

Further Reading

- Ramstead, M. J. D. et al. (2019). Answering Schrödinger's question: A free-energy formulation. *Physics of Life Reviews*, 24, 1-16.
- Friston, K. J. et al. (2022). Designing ecosystems of intelligence from first principles. In *Collective Intelligence*. PNAS.
- Constant, A. et al. (2019). A variational approach to niche construction. *Journal of the Royal Society Interface*, 15, 20170685.
- Bruineberg, J. et al. (2021). The emperor's new Markov Blankets. *Behavioral and Brain Sciences*, 45, e183.
- Seth, A. K. (2021). *Being You*. Dutton, Ch. 10-11.